

Vantage Fleet OS

Design, Build and Delivery of the Maintenance, Dispatch and Compliance Platform

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Proposed duration	34 weeks, seven phases	Classification	Commercial in confidence

1. Executive summary

We propose to design, build, test and deploy the Vantage Fleet OS platform in full, covering all fifty-seven functional requirements, both client applications, the Partner API and the data importers, across thirty-four weeks in seven accepted phases.

Vantage requires a single system to carry a load from intake through assignment, pre-trip inspection, dispatch, post-trip reconciliation, maintenance triage and compliance review, with the workflow itself held as configuration rather than compiled into the application. The requirement set is large but, with three exceptions, conventional. Those three exceptions carry nearly all of the technical risk in the engagement, and the RFP identifies them plainly in clause 5.2: the data-driven workflow engine, the hours-of-service reconciliation ledger, and the eligibility rules that must return identical answers on the server and inside the driver's mobile application.

Our delivery plan is organised around retiring those three risks in the first half of the schedule rather than distributing them across it. The workflow engine and audit trail are complete and accepted by week ten. Eligibility, outcome routing and the mobile foundation follow by week seventeen. The hours-of-service ledger, which is the requirement with the strongest correctness obligation, is built against a frozen configuration schema rather than a moving one.

PROTOTYPE EVIDENCE

We have built a working prototype of the three high-risk mechanics directly against this specification. It implements the configurable status and transition engine with flag-based column resolution, the four-state eligibility determination, a re-derived hours-of-service ledger, inspection outcome routing with override capture, and the immutable audit trail carrying eligibility snapshots. It is covered by thirty-three service-level tests and a transport-level suite that includes cross-company isolation checks. The estimates, sequencing and risk positions in this proposal are drawn from that build rather than from analogy, and the prototype is available for demonstration during evaluation.

Two findings from that work shape the proposal. First, resolving workbench columns by status flag rather than by status name removes risk R-3 by construction instead of by documentation, and costs nothing when done from the outset. Second, the reliable answer to risk R-2, drift between the mobile and server eligibility implementations, is not shared test fixtures but a single compiled rules package

that both runtimes import. That is only achievable if the server and the mobile application share a language, which is the principal reason for the stack we propose in section 4.

2. Understanding of requirements

The platform is a company-scoped multi-tenant application with six roles and a platform-administrator flag layered above them. Owner-operator partners must see only their own loads, vehicles and documents, and that restriction has to hold at the data access layer rather than in the interface, across dashboards, navigation counts and help visibility alike. We read clause 4.1 as requiring capability rules derived from role rather than a fixed permission list per screen, and we have designed accordingly: authorisation is expressed as named capabilities evaluated server-side, so a rule such as dispatch-write can be granted to two roles and a super-user flag without being restated in every route.

The workflow engine is the structural centre of the system. Statuses carry position, behavioural flags, role-visibility arrays and field sets as data, and transitions carry an optional outcome trigger and a single default target. Every board, list column, permission to advance and audit entry derives from those two tables. The consequence, which we treat as a design obligation rather than a side effect, is that no screen may depend on a status name or code. The RFP notes under clause 4.11 that the reference implementation violates this in the workbenches; our implementation does not.

The hours-of-service ledger is the requirement with the least tolerance for error. Clause 4.14 states the invariant precisely: one reconciled ledger per driver per cycle, accumulated across all loads within a route, cleared at each qualifying reset, and attributed against the applicable ruleset resolved in priority order. We treat availability as a derived quantity that is recomputed on every read and never cached, and we test the invariant with generated duty sequences rather than fixed examples.

3. Scope of work

3.1 Functional scope

The following table maps every functional requirement group in clause 4 of the RFP to the phase in which it is delivered and accepted. A complete point-by-point compliance matrix for FR-1 to FR-57 appears in section 14.

TABLE 1. FUNCTIONAL SCOPE BY DELIVERY PHASE

REQUIREMENTS	CAPABILITY AREA AND DELIVERED CONTENT	PHASE
FR-1 to FR-7	Access control, authentication and users. Six roles with a platform-administrator flag; company scoping and capability checks enforced at the query layer; impersonation with anti-escalation safeguards, session badge and exit control; invitation onboarding with resend; searchable user administration; device-bound refresh tokens for mobile sessions.	P1
FR-8 to FR-16	Records and reference catalogue. Carrier companies with expiry thresholds; vehicle records with status and telematics source marker; driver records with ruleset assignment and computed eligibility; vehicle types and maintenance-interval templates; fault and defect catalogues with per-type exclusions; carrier aliases and legacy mappings.	P1
FR-17 to FR-21	Loads and documents. Load entity with the VFO reference format; status-gated list with per-status columns, route grouping, persistent bulk selection and a fixed action bar; detail view with document gallery and per-stage dwell timeline; the created-status edit and delete lock.	P1 to P2
FR-23 to FR-25	Workflow engine. Configurable statuses and transitions covering all flag, colour, role-visibility, field-set and column configuration; reorder with auto-resequence; archive and unarchive preserving history; the workflow flowchart; the role-visibility matrix.	P2
FR-30 to FR-35	Advance, revert, bulk operations and audit. Server-revalidated advance; rank-enforced backward-only revert; bulk advance and bulk revert with union field rules; whole-route advance; the immutable audit trail carrying captured inspection data, eligibility snapshots, stop counts and batch flags.	P2
FR-22, FR-26 to FR-29	Assignment, eligibility and outcome routing. Cascading vehicle-to-driver assignment filtered by eligibility, with a live determination panel and rule-specific help text; ruleset resolution in the specified priority order; inspection outcome computation and auto-routing with mandatory override reason; the shared Record and Advance modal with its fast path.	P3
FR-36 to FR-39	Workbenches, routes and triage. Maintenance and Dispatch boards with faceted cascading filters, live search, optional grouping, bookmarkable URLs and in-place updates; route creation with auto-advance of eligible loads and derived stage badges; per-vehicle fault triage with severity breakdown and out-of-service recomputation.	P3
FR-40 to FR-43	Hours of service and compliance. The per-driver-cycle reconciled ledger with per-entry on-duty override; the compliance workbench with declined-vehicle routing back to maintenance; the compliance record lifecycle and paginated history; generated internal and external audit summaries.	P4
FR-44 to FR-48, FR-52	Bulk upload and integrations. The multi-step standard importer with per-row validation and historical hours verification; the legacy importer with add-alias-and-revalidate; the downloadable template and upload history; the telematics web view and ingestion paths; the bearer-token Partner API and its in-app documentation.	P5

REQUIREMENTS	CAPABILITY AREA AND DELIVERED CONTENT	PHASE
FR-49, FR-51, FR-53 to FR-55	Self-service and administration. The dashboard builder with nine widget types and six chart types on a twelve-column grid; the help centre and article administration; the per-role navigation builder with its non-widening resolver; notification templates with merge fields; the settings store.	P6
FR-5, FR-27	Mobile driver application. Assignment confirmation, pre-trip and post-trip inspection capture and duty-status entry; offline-first storage with conflict-safe synchronisation on reconnect; device-bound tokens and remote wipe.	P3 to P6

3.2 Exclusions and optional items

We have scoped to the exclusions stated in clause 11.2 and exclude accordingly: billing and invoicing logic, with mileage and stop data exported only; business intelligence and data-warehouse dashboards; internationalisation; payment processing; any public third-party interface beyond the telematics Partner API; and automated transmission of compliance summaries to a regulatory portal, which are generated for export and copy-paste only. Two further items are carried as the RFP defines them rather than built out. FR-50 ships as the sidebar-aware reporting placeholder that clause 4.19 specifies, and FR-56 fuel analytics is excluded from core scope and offered as a follow-on item. The temporary maintenance tooling of FR-57 is built for staging use and removed from the production image at build time rather than hidden behind a runtime flag.

4. Technology stack

4.1 Proposed stack

Clause 5.1 permits an equivalent stack with justification and names a TypeScript alternative as acceptable. We propose TypeScript across the server, the web client and the mobile application, on PostgreSQL.

TABLE 2. PROPOSED TECHNOLOGY STACK

LAYER	TECHNOLOGY	BASIS FOR SELECTION
Language and runtime	TypeScript on Node 22 LTS	One language across all three runtimes, which is what makes the shared rules package possible
Server	Express with a thin services layer	Matches the services-layer architecture described in clause 5.2
Database	PostgreSQL 16	As specified; partial unique indexes are used to enforce workflow invariants
Schema and migrations	Prisma	Typed schema, reviewable forward-only migrations applied on release
Web client	React with Vite	Reactive interface with in-place board updates and no full-page reloads
Mobile application	React Native, iOS and Android	Mandatory under clause 5.2; shares the rules package with the server
Mobile offline store	SQLite with an append-only sync queue	Conflict-safe capture for inspections and duty status under NFR-6
Background jobs	BullMQ on Redis	Import processing, notification dispatch, scheduled recomputation
Object storage	AWS S3	Load and inspection documents, per clause 5.4
Authentication	Argon2id hashing, HttpOnly cookie sessions, device-bound refresh tokens	Clause 5.5 password and session requirements
Charting	Chart.js	The six chart types required by FR-49
Spreadsheet handling	ExcelJS with macro execution disabled	Safe parsing for FR-44 to FR-46
Containerisation	Docker multi-stage, non-root, health-checked	Clause 5.3 image requirements
Orchestration	Kubernetes with rolling deploys	Zero-downtime releases with migrations on release
Continuous integration	GitHub Actions	Runs on every push and pull request
Test tooling	Vitest, Supertest, Playwright, Detox	Unit, controller, end-to-end and mobile layers of clause 5.6
Security scanning	Dependency audit and CodeQL in CI	Gated at zero high or critical findings
Observability	Structured JSON logging, OpenTelemetry traces	Clause 5.3 structured logging

4.2 Justification for departure from the reference stack

The reference implementation is Django, and we do not claim the functional set is easier to build in Node. The RFP is correct that the workflow engine, the ledger and the dual eligibility rules carry substantial effort regardless of stack. Our justification rests on one specific point rather than on general preference.

FR-27 requires the eligibility logic to operate consistently on the server and in the mobile application, and risk R-2 records drift between those implementations as medium-likelihood and high-impact, with a single source of truth stated as the preferred mitigation. With a Python server and a JavaScript mobile application, a single source of truth is not attainable; the mitigation necessarily degrades to shared fixtures testing two independent implementations, which detects drift after it occurs rather than preventing it. With one language across both runtimes, the eligibility and hours-of-service rules compile to a single package that the server imports directly and the mobile application bundles. There is then no second implementation that can drift, and the highest-impact risk in the register is closed by construction rather than monitored.

We note the trade-off honestly. If Vantage holds or intends to recruit Python engineering capability for long-term ownership, that is a legitimate reason to prefer the reference stack, and we are equally able to deliver on Django. In that case R-2 reverts to a monitored risk mitigated by shared fixtures, and we would propose a contract-test suite executed against both implementations on every build.

4.3 Forward technology roadmap

The following are outside core scope and are not priced or scheduled here. They are recorded so that the architecture delivered at go-live does not preclude them, and each is reachable without structural rework.

TABLE 3. POST-LAUNCH TECHNOLOGY ROADMAP

CANDIDATE	TECHNOLOGY DIRECTION	ENABLED BY
Reporting engine	Query builder over a PostgreSQL read replica, replacing the FR-50 placeholder	Sidebar auto-listing and the settings store are built to accept it
Regulatory transmission	Portal submission adapters for the compliance summaries excluded under clause 11.2	Summaries are already generated from structured records, not free text
Additional telematics providers	Further adapters behind the existing normalisation seam	The seam is delivered in P5 specifically to make this additive
Business intelligence	Change-data-capture into a columnar warehouse	Immutable audit rows give a clean event stream to replicate
Predictive maintenance	Interval forecasting from accumulated telematics history	Telematics events are retained rather than folded into vehicle counters
Route optimisation and mapping	Mapbox or equivalent over the existing route and stop entities	Routes carry planned stops and target departure from delivery
Enterprise identity	SAML or OIDC single sign-on alongside password authentication	Authentication is isolated behind a single session-issuing service
Fuel analytics	The FR-56 efficiency model, wired to route and telematics data	Offered as a follow-on item as clause 4.25 permits

5. System architecture

The architecture separates a thin transport and authorisation layer from a services layer that holds the domain logic, with the workflow engine driven entirely by configuration rows. The shaded band in Figure 1 is the shared rules package described in section 4.2.

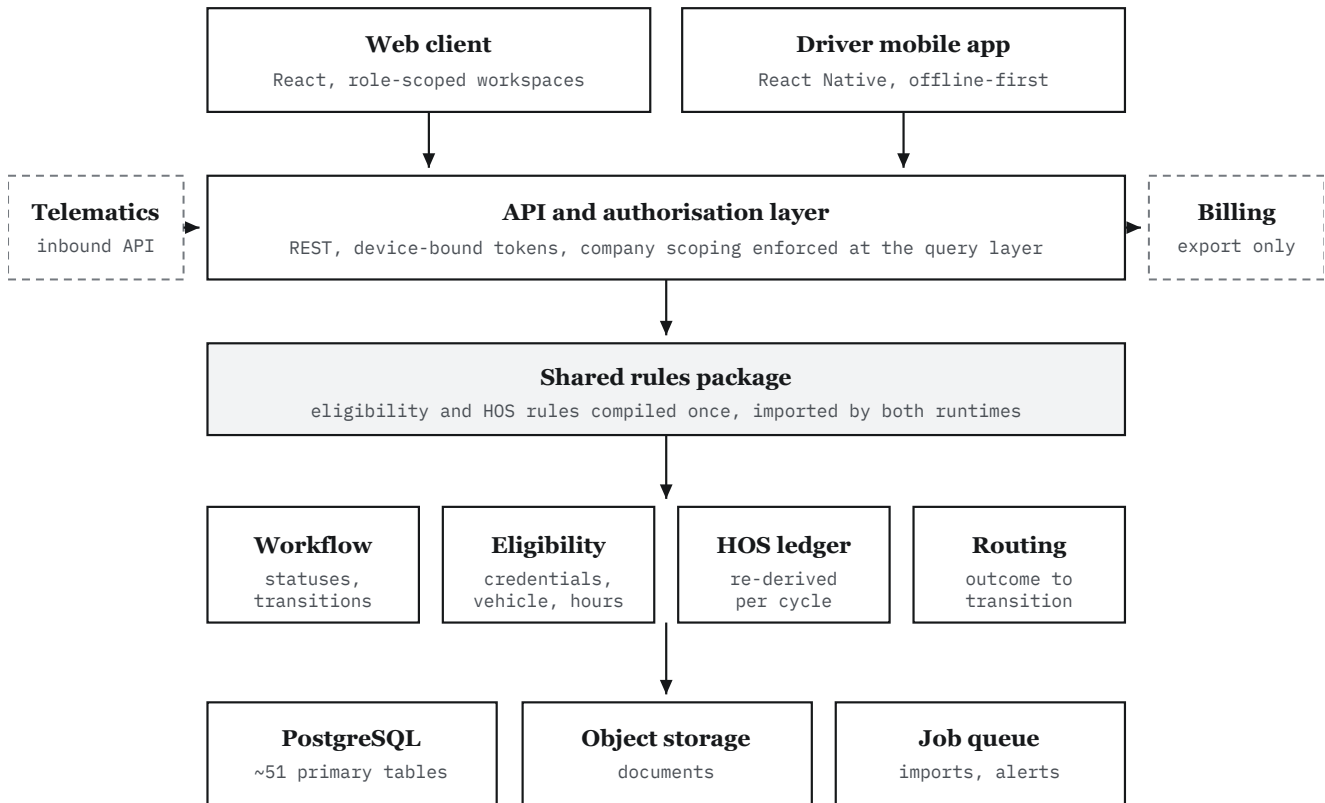


Figure 1. Layered architecture. The mobile application bundles the same rules package the server imports, so an eligibility determination made in the cab matches the one the server produces when the device synchronises. The server remains authoritative and revalidates every submission.

5.1 Data model

Approximately fifty-one primary tables across six domains, consistent with Appendix A of the RFP: loads and workflow, comprising loads, status logs, dispatch statuses, transitions and the defect and fault catalogues; the reference catalogue and legacy mappings; telematics events and duty-status changes; compliance records, routes and carrier companies; configuration and content; and framework tables for storage, jobs and cache. Two rules hold throughout the model. Derived values are never stored, so route stage, driver eligibility and hours availability are computed on read and cannot go stale. Captured audit data is held as a single versioned JSON column on the status log rather than as a set of nullable snapshot foreign keys, which keeps each audit row immutable and self-contained.

5.2 Dispatch workflow and outcome routing

Figure 2 shows the default configured workflow. Every box is a row in the status table and every arrow a row in the transition table, so the shape shown is configuration rather than code and an

administrator can alter it without a release.

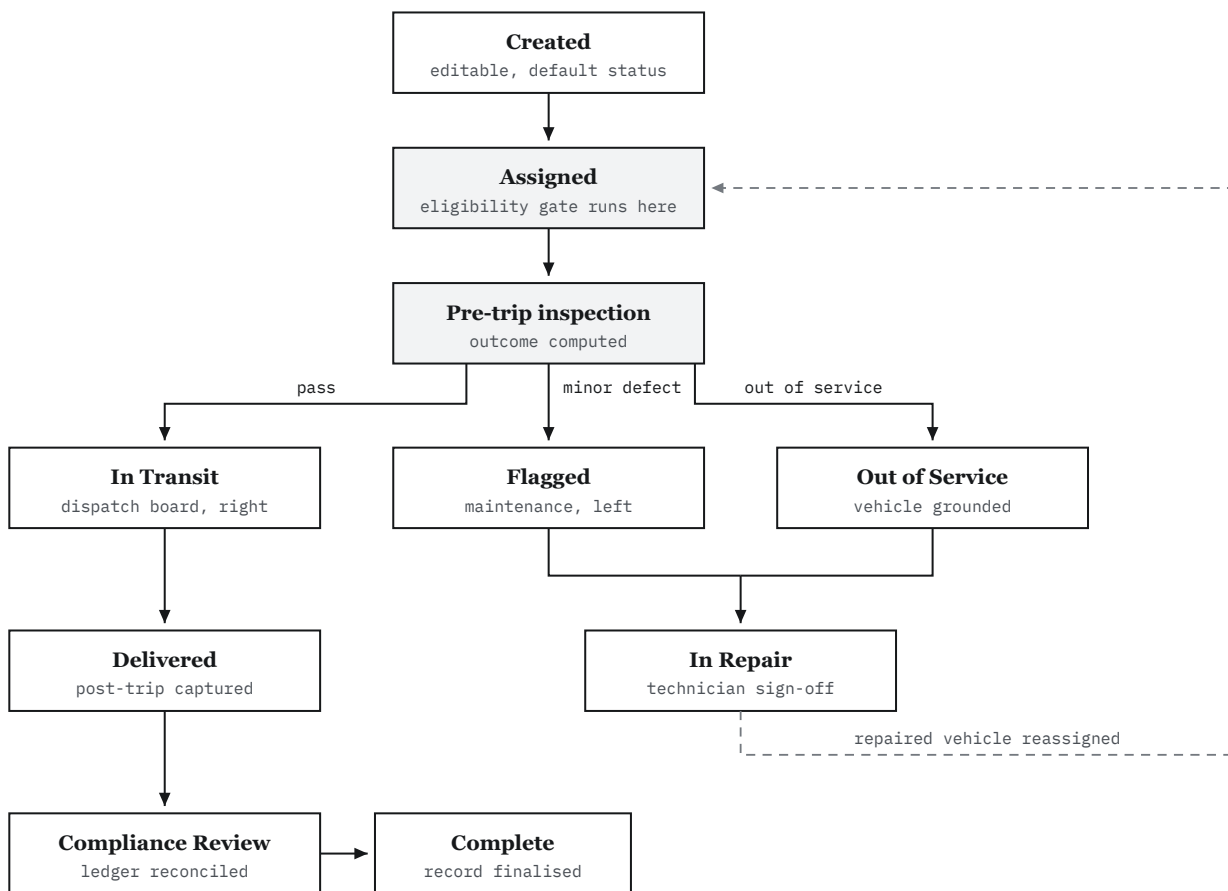


Figure 2. Default dispatch workflow. Solid arrows are configured transitions and the dashed arrow is the repaired-vehicle return path. The three labelled branches leaving the inspection stage are selected automatically by the computed inspection outcome.

FR-28 specifies the outcome priority exactly, and we implement it as a pure function so that it is testable without a database: any out-of-service defect makes the overall outcome out-of-service; failing that, any minor defect makes it a minor defect; otherwise the inspection passes. The computed outcome selects the transition whose trigger matches it. A manual override bypasses that selection and requires a reason, which is validated before any write occurs, so an override without a reason cannot leave a partial record behind. Submitting an inspection and advancing the load occur within one database transaction, leaving no window in which an inspection exists without the status change it caused, or the reverse.

Workbench columns resolve by status flag only. Renaming a status in the workflow configuration screen therefore cannot break the Dispatch or Maintenance board, and we include that rename as an automated regression test rather than as a documented caveat, which is the improvement invited by the note under clause 4.11.

6. Delivery approach

6.1 Method and cadence

Each phase concludes with a deployed increment on an agreed environment, a passing continuous integration run, and a demonstration against the specific requirements that phase claims, which is the acceptance definition given in clause 6.3. Within a phase we work in one-week iterations with a mid-phase review, so that a misread requirement surfaces within days rather than at the handover checkpoint. Requirements are tracked by FR number from proposal through to acceptance, so that at any point in delivery the status of each of the fifty-seven items is visible to both parties.

Scope control carries more weight here than on a typical engagement because clause 11.3 records creep on the configurable engine as the only risk rated high for both likelihood and impact. Our control is a configuration schema freeze at the end of phase two: after that checkpoint, additions to the status, transition or field-set schema are handled as change control rather than as refinement. A configurable system invites indefinite configurability, and this freeze is what prevents phase two from consuming phase four.

6.2 Treatment of the three high-risk mechanics

- **Workflow engine.** Statuses carry an explicit position, and direction is enforced by rank rather than by which edges happen to be configured: advance may move only to an equal or later position, revert only to an earlier one. This makes the backward-only rule of FR-31 structural. Two partial unique indexes make ambiguous auto-routing impossible at the database level, permitting one default target per source status and one transition per outcome per source status.
- **Hours-of-service ledger.** Availability is re-derived from duty-status entries on every read and never cached, as risk R-6 requires. The single-reconciled-ledger invariant is expressed as a property test over generated duty sequences, including reset boundaries and overlapping assignments, rather than as example-based cases alone.
- **Dual eligibility.** One compiled rules package imported by both runtimes, as set out in section 4.2. The mobile application calls the same function the server calls; the server remains authoritative and revalidates on submission.

6.3 Resourcing plan

TABLE 4. TEAM COMPOSITION AND ALLOCATION

ROLE	ALLOCATION	ENGAGED	PRINCIPAL RESPONSIBILITY
Technical lead and architect	1.0 FTE	P1 to P7	Workflow engine, ledger design, architectural decisions, code review
Senior backend engineers (2)	2.0 FTE	P1 to P7	Services layer, Partner API, importers, data model
Frontend engineers	1.5 FTE	P1 to P7	Workbenches, load views, dashboard builder, administration screens
Mobile engineer	1.0 FTE	P3 to P7	React Native application, offline capture and synchronisation
QA and test automation	1.0 FTE	P2 to P7	End-to-end suites, mobile offline scenarios, regression coverage
DevOps engineer	0.5 FTE	P1, P7 weighted	Pipeline, container image, orchestration, observability, release process
Delivery manager	0.5 FTE	P1 to P7	Phase acceptance, change control, Vantage liaison, reporting

Peak allocation is seven and a half full-time equivalents during phases three and four. The technical lead is continuous across the engagement rather than rotated, which matters because the workflow engine and the ledger are decisions taken early whose consequences are felt in every later phase.

6.4 Testing and continuous integration

Continuous integration runs on every push and pull request and gates merge. Unit tests cover the eligibility rules, hours allocation, outcome routing and validation logic. Controller tests cover actions, authorisation gates and company scoping, and include a cross-company isolation case for every domain entity rather than for loads alone. End-to-end tests cover login, the full dispatch lifecycle, the advance modal across both the server and mobile eligibility paths, bulk upload, triage and compliance review. Mobile tests cover offline capture, synchronisation conflict resolution and device-loss handling. Dependency and code scanning are gated at zero high or critical findings, and the linter is gated at zero offences. As clause 5.6 anticipates, the duplicated client and server eligibility logic and the configuration-driven workflow receive disproportionate system-test coverage; in our case the shared rules package reduces the surface that must be tested twice to the call sites rather than the logic.

7. Milestones and timeline

We propose thirty-four weeks, within the twenty-four to forty week band indicated in clause 7. Phases run in sequence because each depends materially on its predecessor, with the mobile application running as a parallel stream from week eleven once the workflow engine is stable enough to build against.

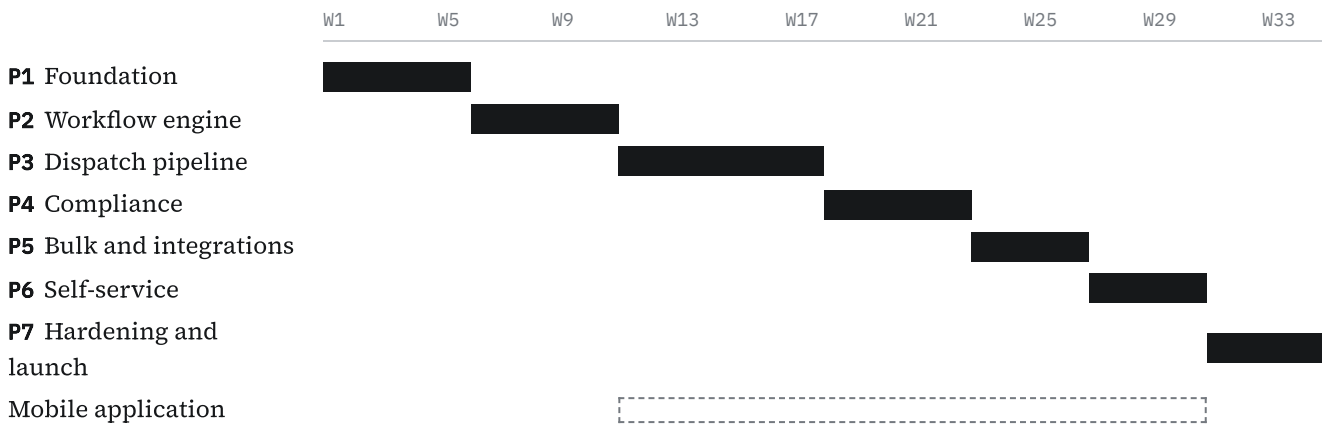


TABLE 5. PHASE DELIVERABLES AND ACCEPTANCE

PHASE	WEEKS	HANDOVER INCREMENT AND ACCEPTANCE DEMONSTRATION
P1	1 to 5	Foundation. Project and pipeline setup, container image and deployment; authentication, users, invitations, impersonation and role-based access; carriers, vehicles, drivers and the reference catalogue; basic load records. <i>Accepted when a scoped user cannot reach another company's data through any route, demonstrated at the query layer rather than the interface.</i>
P2	6 to 10	Workflow engine. Configurable statuses and transitions; advance and revert with bulk variants; the immutable audit log; the workflow diagram and role-visibility matrix. <i>Accepted when a status and transition added through the interface change load behaviour with no deployment. The configuration schema freezes at this checkpoint.</i>
P3	11 to 17	Dispatch pipeline. Assignment flow with live eligibility determination; routes; the Maintenance and Dispatch workbenches; the shared Record and Advance modal; the mobile application foundation. <i>Accepted when identical eligibility inputs return identical verdicts from the server API and from the mobile bundle.</i>
P4	18 to 22	Compliance. The hours-of-service ledger and allocation; the compliance workbench and record lifecycle; generated compliance summaries. <i>Accepted when cumulative on-duty totals reconcile against a driver's declared logs across a full cycle including qualifying resets.</i>
P5	23 to 26	Bulk and integrations. Standard and legacy importers; telematics verification and web view; the Partner API and its documentation. <i>Accepted when a five-hundred-row import validates in under ten seconds with leading zeros preserved and descriptive per-row errors.</i>
P6	27 to 30	Self-service and administration. Dashboard builder; per-role navigation configuration; help centre and articles; notification templates; settings. <i>Accepted when navigation configuration can reorder, rename and hide items for a role but cannot grant that role any access it does not already hold.</i>
P7	31 to 34	Hardening and launch. Security and performance hardening; complete end-to-end coverage including mobile offline synchronisation; documentation; production deployment; user acceptance testing; knowledge transfer. <i>Accepted when NFR-1 to NFR-7 are met under representative load and a zero-downtime rolling deployment with migrations on release is demonstrated.</i>

7.1 Optional leaner release

As clause 6.4 invites, we can additionally deliver a leaner first release comprising phases one to three together with a single-ruleset compliance review, reaching a usable dispatch and maintenance system in approximately nineteen weeks. Dashboards, navigation configuration, the help centre, the legacy importer and the Partner API are deferred in that option, and per-carrier ruleset overrides are excluded. This is offered as an alternative sequencing rather than a reduction in eventual scope, and no pricing is attached to it here.

8. Non-functional requirements

TABLE 6. RESPONSE TO NFR-1 TO NFR-7

REF	REQUIREMENT	HOW IT IS MET AND VERIFIED
NFR-1	Performance budgets	Board and list queries are written against covering indexes with eager loading, so no list or board page issues per-row queries. Budgets are asserted in CI against a seeded dataset, so a regression fails the build rather than reaching staging.
NFR-2	Concurrency	Stateless web workers behind a load balancer, horizontally scaled; sessions held in Redis rather than in process. Load tested at the stated eighty internal and two hundred and fifty mobile sessions during phase seven.
NFR-3	Availability and backup	Rolling zero-downtime deployments with migrations applied on release; daily automated database backups with a documented and rehearsed restore procedure.
NFR-4	Browser support	Latest two versions of Chrome, Firefox, Safari and Edge, verified in the end-to-end matrix; responsive layouts down to tablet width for workbench use.
NFR-5	Accessibility	WCAG 2.1 AA contrast, keyboard-navigable primary flows and labelled form controls, checked by automated audit in CI and by manual keyboard traversal of the dispatch and inspection paths.
NFR-6	Mobile offline	Inspection and duty-status capture write to a local append-only queue with client-generated identifiers, making synchronisation idempotent; reconnection replays the queue with no data loss.
NFR-7	Auditability	Every status change writes one immutable log row carrying actor, timestamps, from and to status, captured inspection data, eligibility snapshot, stop count and batch or reverted flags. Audit rows are never updated or deleted by application code.

9. Security

Authorisation is enforced in the services and query layer, never in the interface alone, and company scoping is applied as a query predicate so that a missing filter fails closed. Passwords are hashed with Argon2id at a tuned work factor, and reset tokens expire within the six-hour window suggested in clause 4.2. Session cookies are HttpOnly, Secure and SameSite, cross-site request forgery protection is enabled everywhere except the Partner API namespace, and TLS is forced in production. Partner API tokens are held per partner in encrypted configuration and compared in constant time. Secrets live in encrypted configuration and appear in neither source nor logs. All input is validated server-side, queries are parameterised through the ORM, spreadsheet parsing runs with macro execution disabled and upload types are validated on content rather than extension. On mobile, offline data is held in encrypted local storage behind device-bound tokens with remote wipe available for lost devices. Dependency and static code scanning run in the pipeline and gate merge.

10. Risk register response

TABLE 7. RESPONSE TO THE RISK REGISTER AT CLAUSE 11.3

REF	RISK	OUR POSITION
R-1	Telematics API changes break ingestion	A normalisation seam sits ahead of ingestion so a provider payload change is absorbed by an adapter without altering the ingestion path. Payload shapes are pinned and contract-tested against the Vantage-provided sandbox.
R-2	Mobile and server eligibility logic drift	Eliminated rather than mitigated. One compiled rules package is imported by both runtimes, which is the reason for the stack proposed in section 4. Shared fixtures still execute against both call sites as a guard.
R-3	Renamed statuses break the workbenches	Columns resolve by status flag and never by name, with a rename regression test in the suite. There is no name dependency to document.
R-4	Scope creep on the workflow and eligibility engine	The configuration schema freezes at the phase two handover, after which schema additions are change control rather than refinement. This is the highest-rated risk in the register and the schedule depends on the freeze holding.
R-5	Spreadsheet import edge cases	Per-row validation with descriptive errors and leading-zero normalisation runs before any write. Malformed files fail at parse with a usable message rather than part-way through a transaction.
R-6	Hours-of-service reconciliation errors	Availability is always re-derived server-side and never cached. The single-reconciled-ledger invariant is a property test over generated duty sequences covering reset boundaries and overlapping assignments.
R-7	Data migration from the legacy system	A dedicated migration rehearsal in phase five runs the legacy importer against a full production extract before cutover, so alias gaps and mapping misses surface against real data rather than samples.
R-8	Mobile offline synchronisation conflicts	The conflict rule is settled in phase three before offline capture is built. Inspection and duty-status entries are append-only with client-generated identifiers, making synchronisation idempotent and conflicts structurally rare; field testing in phase seven covers the remainder.

11. Support, warranty, intellectual property and handover

We offer a defect-fix warranty of sixty days from production go-live, exceeding the thirty-day minimum required by clause 10.1, during which any failure to meet the specification is remedied at no additional cost. Our proposed ongoing support offering, and the severity definitions and response and resolution targets attached to it, are set out below and are open to alignment with the hours and targets Vantage intends to complete in clause 10.2.

TABLE 8. PROPOSED SUPPORT RESPONSE TARGETS

SEVERITY	DEFINITION	RESPONSE	RESOLUTION TARGET
S1	Production unavailable, or dispatch and compliance recording blocked for all users	1 hour	8 business hours
S2	Major function unavailable with no workaround; single role or workbench blocked	4 business hours	2 business days
S3	Function impaired with an available workaround	1 business day	10 business days
S4	Cosmetic defect, documentation correction or minor request	3 business days	Next scheduled release

All source code, deliverables, documentation and associated intellectual property created under this engagement are owned by Vantage and are developed in a Vantage-controlled repository from the first commit rather than transferred at completion. Every third-party and open-source component is disclosed with its licence, and we will use no component whose licence terms are incompatible with Vantage's ownership, modification and commercial hosting rights. We incorporate no pre-existing vendor intellectual property that is not licensed to Vantage on a perpetual, royalty-free basis.

Documentation delivered at handover comprises architecture and data-model documentation, deployment and operational runbooks, Partner API documentation, and an administrator configuration guide covering the workflow engine, the hours-of-service ledger and the settings store. Handover concludes with a knowledge-transfer programme for Vantage staff or a nominated maintenance partner, run as working sessions against the live system rather than as a presentation.

12. Assumptions and dependencies

Consistent with clause 11.1, this proposal assumes that Vantage provides telematics sandbox access for Partner API integration testing; the SMS gateway and an object storage account; reference and seed data covering vehicle types, fault codes, defect categories, statuses, initial users, carrier aliases and legacy mappings; and a single production hosting environment unless we are separately engaged for managed hosting. We assume a purchased administration theme is acceptable and that bespoke visual design is a follow-on item, and that hours-of-service rulesets are expressed per applicable federal exemption category. We assume Vantage makes staff available for phase reviews and user acceptance testing within one week of each handover; where that availability slips, the affected phase boundary moves rather than the subsequent phase compressing. We treat the functional requirements in the RFP as the authoritative reference in the event of any conflict with the reference implementation.

13. Vendor particulars and qualifications

The RFP requires evidence against VQ-1 to VQ-10 together with company particulars and contactable references. Those items are specific to the submitting entity and are to be completed before submission; they are not asserted here.

To be completed before submission. Legal entity name, registration and trading history; team size; professional indemnity and public liability insurance details (VQ-10). Case-study evidence against VQ-1 to VQ-8, covering multi-tenant SaaS delivery, data-driven workflow engines, regulated compliance workflows, bearer-token API integrations, bulk ingestion, stack proficiency, testing and continuous integration discipline, and offline-first mobile delivery. At least two contactable client references (VQ-9). Signed statement of ninety-day validity.

One qualification item can be evidenced directly rather than by reference. Against VQ-2, experience building configurable workflow engines, and VQ-3, regulated compliance workflows with strong correctness guarantees, we can demonstrate the working prototype described in section 1, built against this specification, including its test suite and its handling of the flag-based column resolution that clause 4.11 identifies as a weakness in the reference implementation.

14. Functional requirement compliance matrix

Compliance is stated as Full where the requirement is delivered as specified, Alternative where we propose a different means to the same end, and Optional where the RFP itself marks the item optional or excluded from core scope.

TABLE 9. POINT-BY-POINT COMPLIANCE, FR-1 TO FR-57

REF	REQUIREMENT	COMPLIANCE	REF	REQUIREMENT	COMPLIANCE
FR-1	Role model	Full	FR-30	Advance	Full
FR-2	Authorisation enforcement	Full	FR-31	Revert	Full
FR-3	Company scoping	Full	FR-32	Bulk advance	Full
FR-4	Impersonation	Full	FR-33	Bulk revert	Full
FR-5	Authentication	Full	FR-34	Whole-route advance	Full
FR-6	Invitation onboarding	Full	FR-35	Immutable audit trail	Full
FR-7	User administration	Full	FR-36	Maintenance Workbench	Full
FR-8	Carrier companies	Full	FR-37	Dispatch Board	Full
FR-9	Vehicles	Full	FR-38	Routes	Full
FR-10	Driver records	Full	FR-39	Triage workbench	Full
FR-11	Vehicle types	Full	FR-40	Hours-of-service allocation	Full
FR-12	Maintenance interval templates	Full	FR-41	Compliance Workbench	Full
FR-13	Fault and DTC catalogue	Full	FR-42	Compliance record lifecycle	Full
FR-14	Inspection defect categories	Full	FR-43	Compliance summaries	Full
FR-15	Carrier aliases	Full	FR-44	Standard importer	Full
FR-16	Legacy mappings	Full	FR-45	Legacy importer	Full
FR-17	Load entity	Full	FR-46	Template and history	Full
FR-18	Load list view	Full	FR-47	Telematics web view	Full
FR-19	Load detail view	Full	FR-48	Telematics ingestion	Full
FR-20	Edit and lock rule	Full	FR-49	Dashboard builder	Full
FR-21	Documents	Full	FR-50	Reports placeholder	Optional
FR-22	Assignment form	Full	FR-51	Help centre and articles	Full
FR-23	Configurable statuses	Full	FR-52	Partner API	Full
FR-24	Configurable transitions	Full	FR-53	Navigation builder	Full
FR-25	Workflow management UX	Full	FR-54	Notification templates	Full
FR-26	Eligibility determination	Full	FR-55	Settings	Full
FR-27	Dual-implementation constraint	Alternative	FR-56	Fuel analytics	Optional
FR-28	Outcome routing	Full	FR-57	Maintenance tooling	Full
FR-29	Shared advance modal	Full			

Three entries warrant a note. FR-27 is marked Alternative because we exceed the requirement rather than meet it as written: instead of two implementations kept in synchronisation, we deliver one

implementation used by both runtimes. FR-50 and FR-56 are marked Optional because the RFP itself designates them so; FR-50 is delivered as the specified placeholder and FR-56 is offered as a follow-on item. FR-36 and FR-37 are marked Full and additionally resolve workbench columns by flag rather than by name, as invited by the note under clause 4.11.

15. Compliance statements

This response addresses every section of the RFP and follows the response checklist at clause 12. It contains no pricing, rate cards, fixed fees or budget figures, in accordance with clauses 3.9 and 12. The proposal remains valid and open for consideration for ninety days from the submission date. We accept the instructions and conditions set out in the RFP, and we acknowledge that Vantage is under no obligation to award any work.